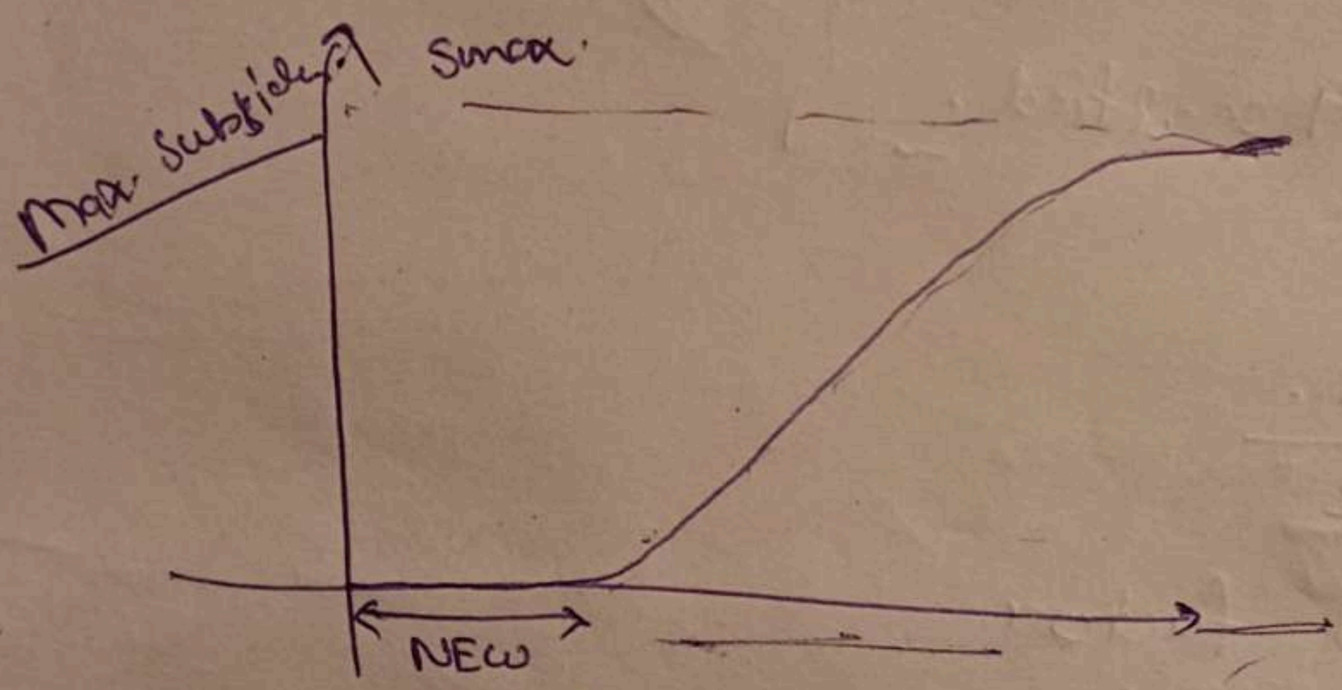
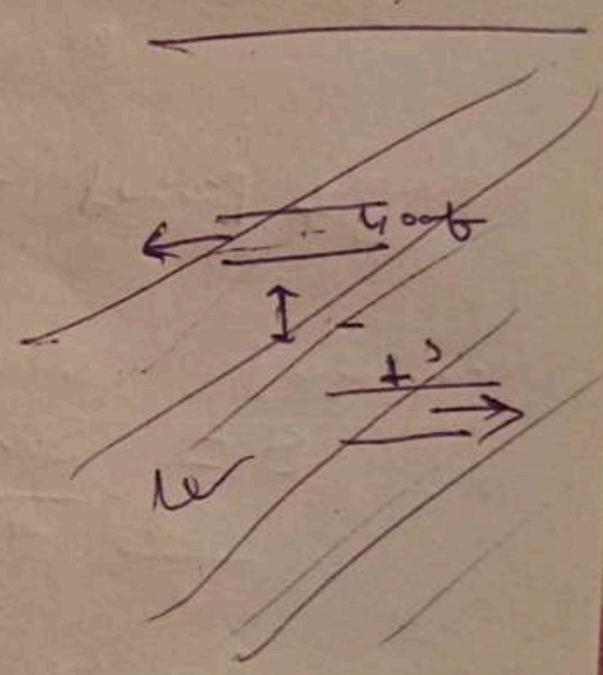
Subsidence is higher in existing goal
- Multi seam
Indian mine
- shallow / moderate cover

H - Depth
h - Mining height
e - extraction % or traction
L - width
S - max. subside

Smax max. possible -
NEW - Non-Effective width to depth ratio
- width upto which no significant subsidence happen.





Panel length (L) / Depth of cover (H) } for single & multi-seam

$\frac{H}{h} e < 0.3$ for discontinuous

$$NEW = \frac{\text{Effective width}}{\text{depth}}$$

width << depth → cannot develop fully max. subsidence

Source vs L/H

$$n = \frac{L/H}{NEW}$$

$$n = \frac{L}{H \times NEW}$$

↳ Normalised Mining cond^{n}

Mine A

$\frac{L}{H} = 0.5$

$NEW = 0.5$

$n = 1$

157 panel – 9 coalfield

$$S = \frac{S_{max.}}{he}$$

S - subsidence factor

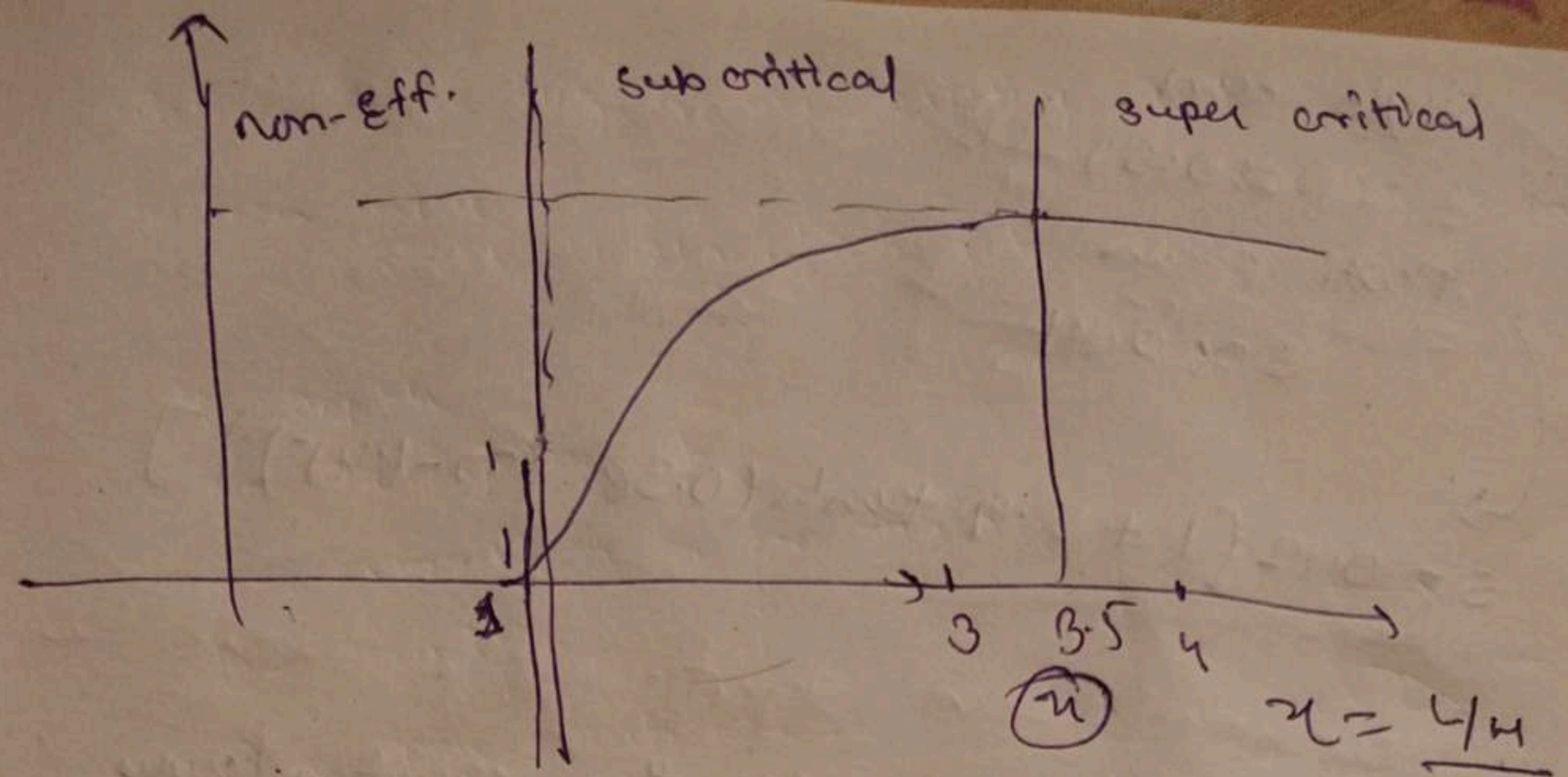
$$S_{max} = S\,he$$

$$n = \frac{L/H}{NEW}$$

style

(1)

(2)



$n < 1 \Rightarrow S \approx 0$

$1 < n < 3.5 \Rightarrow S \approx 0.69$ Transition zone

$n > 3.5 \Rightarrow$ max. subsid. has essentially reached

$\boxed{S = 0.69}$

---

tanh — hyperbolic tangent function.

Single $S = 0.33\{1 + 1.1 \tanh(1.4(n - 1.8))\}$

$n = \frac{L}{H}$ NEW

① when n as small as
tanh as (large negative no) $= -1$

$\boxed{S \approx 0}$

narrow panel → very little max. subsidence

② when n (1 – 3.5)

S ↑ tanh ≈ 1 large +ve.

$S = 0.33[1 + 1.1]$

$= 0.693$

Single seam
$S \simeq 0.69$

Multi seam
$S \propto 0.81$

$S = 0.12\left[1 + 5.9 \tanh\left(0.38\,(n - 1.45)\right)\right]$

R^2 tells us how well the equation/curve fits the actual observed data.

$R^2 = 0.85$ — single-seam
0.86 — multi-seam

Critical width

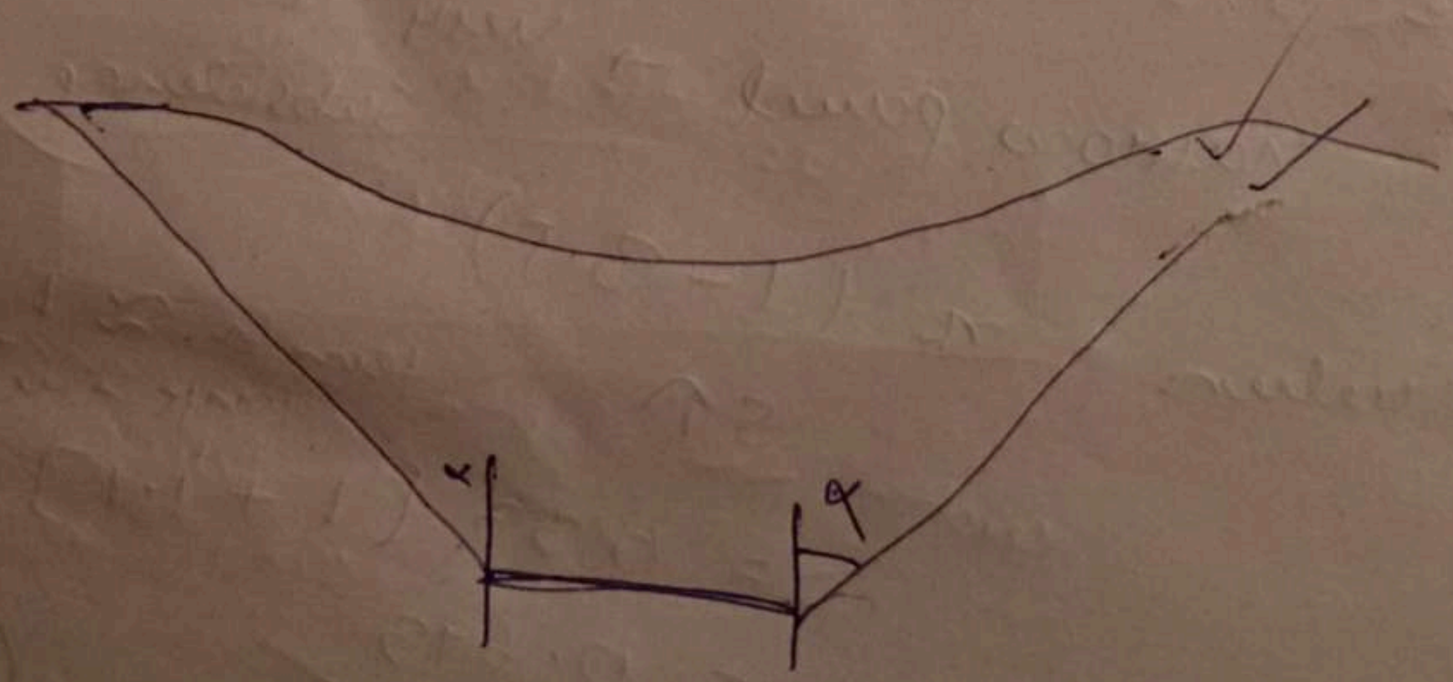
$$\frac{L}{H} = \frac{3.5 \times NEW}{W}$$

for multi seam.

$S = 0.81$

$$S = \frac{S_{max.}}{4e}$$

$$\boxed{S_{max} = 0.81\,4e}$$



Angle of draw → Geometric Property

how much the ground sinks → how much the ground tilts / deform.

Stowing → filling void space
↳ sand
rock
mine waste

Single - seam $S = 0.69$ W

Multi - seam $S = 0.81$ W

But for stowing $S \approx 0.035$

that's drawn smaller

Slope measures the rate at which subsidence changes horizontally,

$$q = \frac{dS}{dx}$$

q - max. slope

S - max. subsidence

H - depth

k_1 = const

$$q = k_1 \frac{S}{H\downarrow}$$ W

Ground deformation can be more steeper for a shallower mine

$E(-) = K_2 \left(\frac{S}{H}\right)$ compressive strain

$E(+) = K_3 \left(\frac{S}{H}\right)$ tensile strain

Climate

~~forme~~

Correct effect can significantly change the draw limit.

G - Slope

E - strain Measure.

Angle of draw

↳ How far mining influence extends sideway



